Kasra Malihi

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Education

University of Copenhagen — Copenhagen, Denmark
Master of Science in Computer Science — Sep. 2025 - present
GPA: 11.67/12

Sharif University of Technology — Tehran, Iran
Bachelor of Science in Computer Engineering — Oct. 2020 - Feb. 2025
GPA: 18.89/20 – Major GPA: 19.33/20

Research Interests

Language Models (esp. Alignment, Post-training)

Reinforcement Learning (RL for LLMs, RL for Reasoning)

Theoretical Machine Learning, Optimization for Deep Learning

Publications

Less Noise, Same Certificate: Retain Sensitivity for Unlearning — Under review for ICML 2026
*Carolin Heinzler, **Kasra Malihi**, Amartya Sanyal*

Achievements

Ranked **6th nationwide** among more than **150,000** participants in Iran National University Entrance Exam (AKA. KONKUR) in Mathematics & Physics Branch, 2020.

Accepted to **ETH Zurich** Computer Science Mobility Program for the 2026/27 academic year, including a 2,000 CHF/Semester SEMP scholarship.

Experience

University of Copenhagen — Research assistant

Language and Multimodal Processing (LAMP) Group — Oct. 2025 - present

Conducting research on Pixel Language Models, supervised by **Prof. Desmond Elliot**. Migrated the PIXEL model from a generative ViT-MAE paradigm to a Headless Contrastive architecture. Explored two structural variants: one retaining a decoder and a decoder-free model utilizing sequence-preserving [MASK] token replacement. Deprecated the pixel-reconstruction head in favor of a Contrastive Weight Tying (CWT) objective over patch embeddings. The contrastive models proved more stable during downstream fine-tuning than the generative baseline.

Foundations of Responsible Machine Learning (FoRML) — Dec. 2025 - present

Conducting research on Machine Unlearning, supervised by **Prof. Amartya Sanyal**. Proposed a data-dependent noise calibration technique that bypasses the conservative global sensitivity limits adapted from Differential Privacy. By leveraging properties of the fixed retain set, such as empirical curvature and margins, this approach maintains unlearning guarantees while reducing required noise and improving model utility. Validated these bounds theoretically and empirically across passive mechanisms (e.g., PCA, SVM, ERM) and active unlearning algorithms (Descent-to-Delete, Newton-step updates). This work is currently under review at ICML 2026. Accepted @ TPDP.

Sharif University of Technology — Research assistant

Machine Learning Lab — Sep. 2024 - Aug. 2025

Conducting research on adversarial attacks and defenses on LLMs, supervised by **Prof. M. Soleymani Baghshah**. Developed an attack-agnostic defense method by training MLP on hidden states of (*request*, *response*) pairs. Demonstrated the impact of layer selection, aggregation function, and token position, and showed that layers with lower Centered Kernel Alignment (CKA) are better candidates for representation reading/engineering. Improved the robustness of the model against adversarial attacks in HarmBench to **84.98%** while keeping the Benign Acceptance Rate (BAR) at **96.4%**. This work was completed as my B.Sc. thesis.

Max Planck Institute for Informatics — Undergraduate summer intern

Artificial Intelligence aided Design and Manufacturing (AIDAM) Group — Jul. 2023 - Sep. 2023

Collaborated with **Prof. Vahid Babaei** on a 3D printing interface, translating joystick and keyboard inputs into G-code for precise printer control. Developed a custom C++ Unreal Engine plugin to enable serial communication within Blueprints.Preprocessed and visualized 3D model slicer outputs as layer-by-layer outlines within the gameplay environment.

Teaching Assistant Experience

Deep Learning† (Prof. M. Soleymani Baghshah) — Spring 2025
Theoretical Machine Learning (Prof. F. Seyyedsalehi) — Spring 2024
Linear Algebra (Prof. HR. Rabiee; Dr. Ramezani; Dr. SH Ghorban) — 2023–2025
Theory of Languages and Automata (Prof. M. Izadi) — 2023–2024

Course Work

University of Copenhagen: Advanced Topics in Machine Learning, Advanced Topics in NLP, Online and Reinforcement Learning

Sharif University of Technology: Machine Learning†, Artificial Intelligence, Convex Optimization, Advanced Information Retrieval, Algorithmic Game Theory

Projects

Compositional Skills of Transformers — Fall 2025

Reproduced results of *"Generalization without Systematicity: On the Compositional Skills of Sequence-to-Sequence Recurrent Networks"* with Vanilla Transformer with teacher forcing. Improved its results by implementing a similar architecture to one proposed in *"Less is more: Recursive reasoning with tiny networks"*.

IMDB Search Engine — Spring 2024

Developed a search engine for a movie database scraped from IMDB, Implementing a multitude of advanced indexing, spell correction, and embedding techniques, with a Collaborative Filtering recommendation system and a conversational retrieval component using RAG.

C-minus Compiler — Spring 2023

Coded a compiler for a simplified version of the C language using Python. Implemented lexer, LL(1) parser and intermediate code generator for said language. A semantic stack was also implemented to handle recursion.

Yu-Gi-Oh! — Spring 2021

Implemented the famous trading card game with a group of colleagues using Java and JavaFX for graphics.

Technical Skills

Programming Languages: Python [PyTorch, TensorFlow, scikit, pandas, transformers, CVXPY ...] (fluent); C/C++, Java[Scala, Kotlin], R

Version Control & Cloud Computing: Git, GitLab, Docker

Miscellaneous: Unreal Engine, SQL, LaTeX

Language Proficiency

Persian: Native

English: Fluent **TOEFL iBT©**: 105 (Reading: 28, Listening: 30, Speaking: 23, Writing: 24)

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[1]†Graduate-level course.